DOCUMENT 1:
INTRODUCTION TO DTAF AI PIPELINE

1. What is DTAF?

While existing frameworks address individual aspects of Digital Twin (DT) implementation, no single approach simultaneously tackles data readiness, simulation capability, maturity assessment, and trust. The Digital Twin Analytical Framework (DTAF) fills this gap by combining complementary analytical frameworks into a cohesive methodology that results in a practical DT implementation roadmap.

2. What are the frameworks that make up DTAF?

The frameworks that make up DTAF are the following. Figure 1 shows they relate to one another.

- **Digital Twin Consortium's (DTC) Capabilities Periodic Table (CPT):** a requirements definition framework to determine the needed DT capabilities [1].
- **Data Element Mapping and Analysis (DEMA):** an interview-driven approach to uncover tribal knowledge and map the flow of data elements for a system or process [2].
- **4R Framework:** a framework to categorize DTs into four increasing levels of capability, enabling organizations to plan DT development to match their desired maturity and functional scope [3][4].
- **4S Framework:** a framework to categorize simulations into four increasing levels of capabilities, guiding the appropriate scope and utility of simulations within the DT [5][6].
- **Digital Twin V (DTV):** a framework to apply iterative verification and validation to evaluate whether the DT is trustworthy and ready to guide decision-making [7].

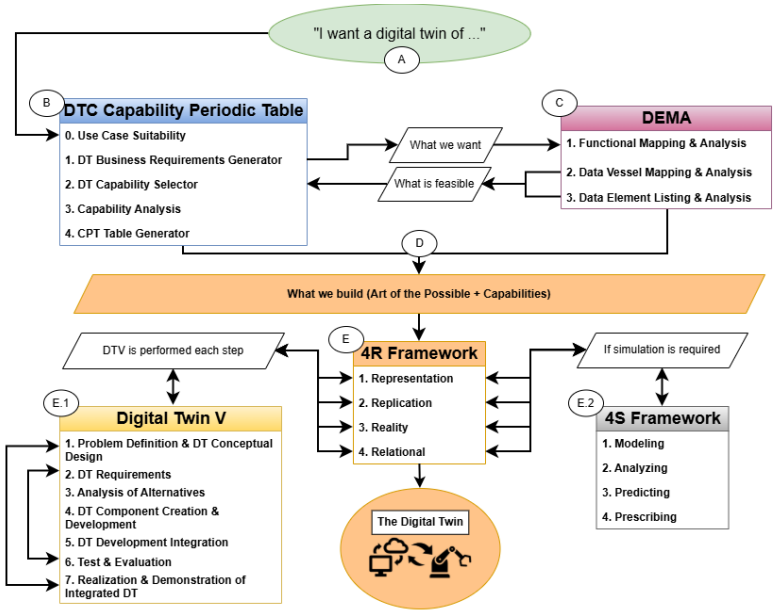


Figure 1: DTAF Structure and Flow.

3. What is the DTAF AI Pipeline?

The DTAF AI Pipeline is a Python and LLM-based tool that facilitates DTAF applications. The DTAF AI Pipeline walks users through a series of simple questions about their proposed DT use case and their current access/format of data. The pipeline then translates the user inputs into DT concepts, references built-in knowledge databases for the DTAF component frameworks, automates the analysis while tailoring guidance to the specific use case, and then concludes by outputting a DT implementation roadmap document to achieve feasible capabilities. The portions of the DTAF AI Pipeline related to the Digital Twin Consortium's Capabilities Periodic Table (DTC CPT) were built from their open-source GitHub repository available on their website [1].

4. Why use the DTAF AI Pipeline?

The DTAF AI Pipeline is designed to simplify the DT development process by leveraging a chatbot-style LLM. Although the DTAF was created specifically for small and medium manufacturers (SMMs), it can be used by any person wanting guidance on where to begin in DT development. SMMs (and many people in various industries) lack the time and resources to become experts in the various terminologies and frameworks surrounding DT design and development. The DTAF AI Pipeline allows users to leverage component frameworks without requiring specialized expertise; by responding to straightforward, guided inquiries in natural language, organizations can concentrate on executing the resulting implementation roadmap to build their DT.

5. What questions does DTAF answer about DT development?

The questions DTAF answers can be grouped and summarized based on the underlying frameworks:

- **DTC CPT:** Which capabilities are essential for my DT use case?
- **DEMA:** Do I have the data elements that I need to achieve my desired DT capabilities?
- **4R:** What level of maturity do I need my DT to achieve?
- **4S:** Do I need simulation to be part of my DT, and if so, what level of capability?
- **DTV:** How do I verify and validate my DT to ensure trustworthiness?

6. What are the outputs of the DTAF AI Pipeline?

The primary output generated is the DT Build Guide. The DT Build Guide consolidates all recommendations and action items produced throughout the AI Pipeline into a single, user-friendly HTML-based report. Several additional support files are also generated and linked within the final .html report for users who require deeper technical analysis. The DT Build Guide contains the actions required to build a DT to the recommended maturity level, achieve the necessary simulation capability level, and establish an appropriate verification and validation strategy to ensure the resulting DT can be trusted for decision-making. Figure 2 shows an example of the

front page of the final output, which includes a high-level breakdown of the recommended DT capabilities, 4R and 4S levels, and overall guidance for DTV.

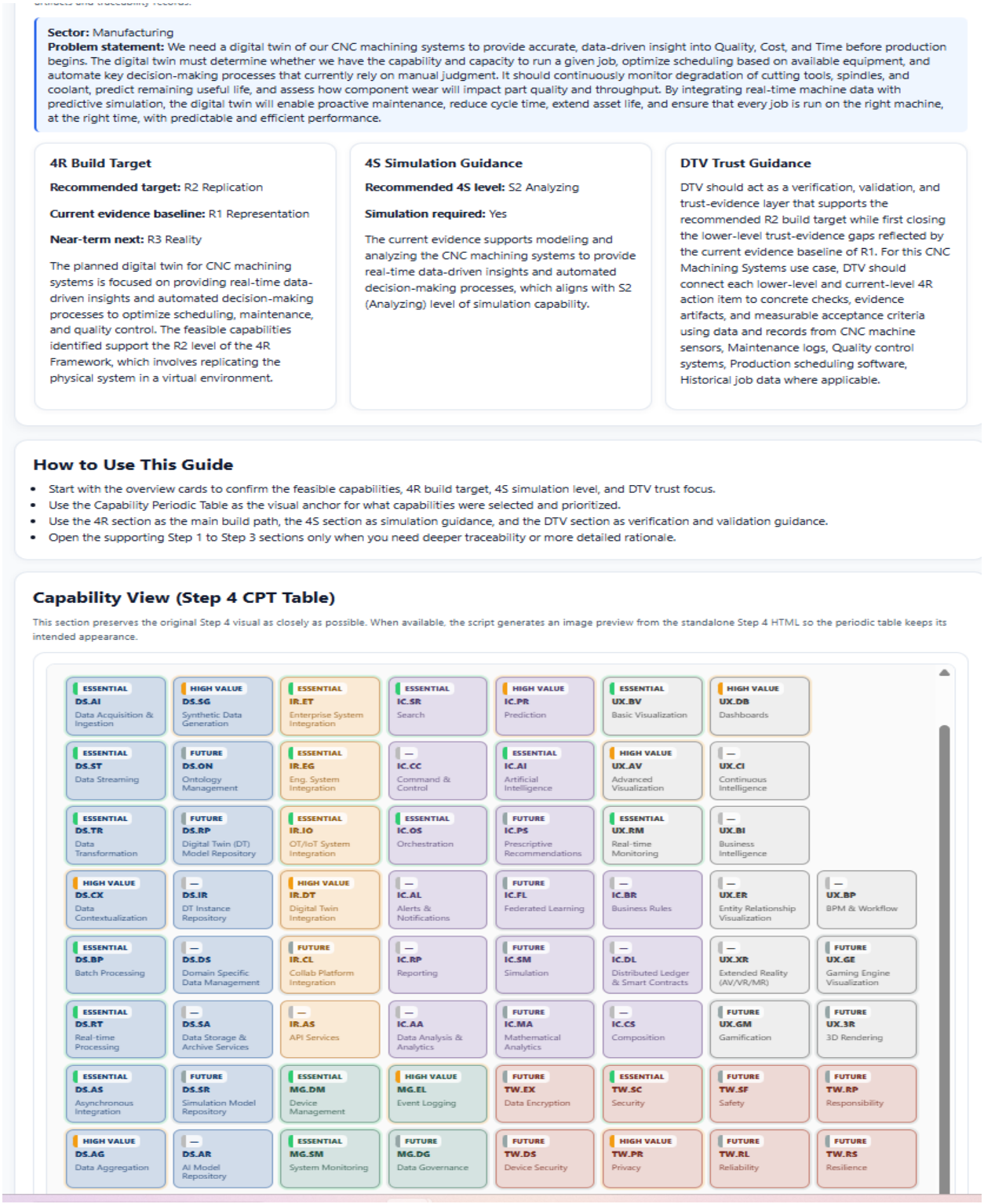


Figure 2: Example DT Build Guide (DTAF AI Pipeline Output).

7. How do you use the DTAF AI Pipeline outputs?

The information from the “Document 2: Quick-Start Guide (How to Download DTAF)” provides what is needed for a user to begin moving from an initial DT idea to a more clearly defined implementation path. As an .html file, it can be saved for users to review, shared with stakeholders, saved for future references, or printed as a planning document. The guide provides recommendations for creating DT design plans, simulation capabilities, and verification and validation plans for the system.

References

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